

**Problem 2**

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**Problem 1** Given the polynomial function  $q$  defined by  $q(v) = 2v^3 - 5v^2 + 7v - 9$  find:

- (a) The slope of the tangent line to the graph of  $q$  at the point where  $v = 3$  using the limit definition of a derivative.
- (b) The slope of the tangent line to the graph of  $q$  at the point where  $v = 3$  using the “short-cut derivative rules” to find a formula for  $q'$  and evaluating  $q'(3)$ .
- (c) The equation of the tangent line to the graph of  $q$  at  $v = 3$ .